

Hierarchical Coordination of Networked-Microgrids Toward Decentralized Operation: A Safe Deep Reinforcement Learning Method

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Abstract—Multiple individual microgrids can be integrated as a networked microgrid system for enhanced technical and economic performance. In this paper, a two-stage data-driven method is proposed to hierarchically coordinate individual microgrids towards decentralized operation in a networked microgrid (NMG) system. The first stage schedules active power outputs of micro-turbines and energy storage systems (ESSs) on an hourly basis for energy balancing and cost minimization, where ESSs are controlled by a local P/SoC droop scheme. In the second stage, the reactive power outputs of PV inverters are dispatched every three minutes based on a Q/V droop controller, aiming to reduce network power losses and regulate the voltage under real-time uncertainties. At offline training stage, a multi-agent deep reinforcement learning model is trained to learn an optimal coordination policy, enhanced by a safety model framework. For online application, the trained agent can work locally in a decentralized manner without information exchanges, and the safety model can also be applied to monitor and guide online actions for safety compliance. Numerical test results validate the effectiveness and advantages of the proposed method.

Index Terms—Networked-microgrid, hierarchical coordination, decentralized operation, deep reinforcement learning, safety model.

I. INTRODUCTION

A TYPICAL microgrid consists of multiple distributed energy resources (DERs), including distributed generators (DGs), energy storage systems (ESSs), and controllable loads [1]. To further improve system's reliability and economy, a networked microgrid (NMG) can be composed by interconnecting multiple individual microgrids within a physically coherent area [2]. For example, Singapore has built a NMG platform, called REIDS, in Semakau Island, which consists of eight microgrids owned by different entities [3]. The NMG system can work

standalone in an islanded mode or be tied to the upstream distribution network as grid-connected mode [4]. However, effective coordination in the NMG becomes more challenging due to more complicated structure and internal interactions, especially under the uncertainties of renewable energy resources (RESs). This paper focuses on active and reactive power management of the NMG system under a grid-connected mode.

Conventionally, centralized energy management aims to minimize the total generation cost or power losses in the system [5], [6]. However, individual microgrid owners in the NMG may have their own targets and profits. Hence, distributed energy management in the NMG is needed, where each individual microgrid only manages its own energy dispatch while coordinating with other microgrids through information exchange. Based on augmented Lagrangian decomposition, the decomposition method is one of common solutions for distributed energy management [7], including alternation direction method of multipliers (ADMM) [8], dual decomposition [9], consensus algorithm [10]. Besides, multi-agent game methods are utilized to solve transactive energy issues in the NMG, by modelling energy management problem into a multi-agent game framework [11], [12]. Nevertheless, distributed energy management requires information exchange among individual microgrids, and the above methods are generally model-based, which require accurate online model and system parameters. Besides, the performance of model-based methods heavily relies on the prediction of renewable energy and load demand, which could deteriorate under high prediction errors.

On the other hand, given the high R/X ratios in the low-voltage distribution network, node voltages are affected by both active and reactive power. Because of the high-level RES penetration and stochastic variations of load demands in the system, voltage would experience fast and uncertain deviations. To realize voltage regulation in the distribution network, voltage/var control (VVC) is normally applied by scheduling reactive power outputs of multiple devices, such as on-load tap changers (OLTCs), capacitor banks (CBs), photovoltaics (PV) inverters [13], [14], [15], [16]. In [13], a multi-timescale coordinated voltage control framework was reported, where OLTCs and CBs are scheduled hourly and the RES inverters are dispatched every 15 minute to handle operational uncertainties from load demand and generation. To achieve better system-wide voltage control performance, a hierarchical framework was designed in [14],

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including three stages: day-ahead, hourly ahead, and real-time. However, existing voltage control methods are model-based way and rely on the centralized structure, which may induce communication burden and cannot address real-time variations of load and RES generation [16]. Besides, most existing control strategies only consider voltage regulation and network loss minimization but ignore other economic issues. In this regard, it is significant to integrate energy management with voltage regulation by simultaneously coordinating active and reactive power resources in the NMG.

In recent years, with significant advancement of artificial intelligence, data-driven methods have attracted considerable attention in the power industry. In particular, reinforcement learning (RL) is capable of solving optimization problems of dynamic systems based on “trial and error” interactions [17]. In [18], a RL-based energy management system (EMS) was proposed to reduce peak-to-average ratio at the demand-side in an NMG. Ref. [19] designed a two-level EMS framework based on RL, which can determine optimal retail electricity prices for energy trading. Authors in [20] developed a resilient EMS in the NMG that maintains normal operation under communication failures based on an improved deep reinforcement learning (DRL) algorithm. For VVC, there are also several publications that report RL-based solutions [21], [22], [23]. In [21], a multi-agent DRL algorithm was designed for real-time control of PV inverters, mitigating fast voltage violation and minimizing total network loss. In [22], authors developed an adversarial RL-based method for voltage control, which aims to improve online safety and efficiency. For most RL algorithms, the conflict between policy’s safety and exploration is a key challenge: in order to learn an optimal solution, more stochastic actions are needed to explore more scenarios during the learning process, which may introduce unsafe actions in the process. In existing works, penalty-based reward [24], [25] and *Lagrangian*-based action-value function [26], [27], [28] are commonly used to avoid constraint violations, which are implemented at offline learning stage. However, such methods are unable to monitor the safety and take actions during online stage if unlearned/unsafe cases occur. Some works report the safety model as an intermediate layer before action execution at online stage [29], [30], but the training of safety models may require the prior knowledge.

To fill the research gaps above, a new data-driven approach for NMG operation is proposed to hierarchically coordinate individual microgrids towards decentralized operation. The contributions of this paper are listed as follows:

- 1) At the first stage, active power setpoints of microturbines (MTs) and active power reference outputs of ESSs are dispatched by *P*-agents every hour. The aim is to minimize the generation cost of each individual microgrid and total network power loss. The input of *P*-agents includes electricity prices, historic State of Charge (SoC) values, and historic PV/load data. For ESS devices, a P/SoC droop-based control strategy is designed, where the ESS output is determined by its SoC and reference output value.
- 2) At the second stage, reactive power reference outputs of PV inverters are dispatched every three minutes to minimize network power loss and eliminate voltage violations.

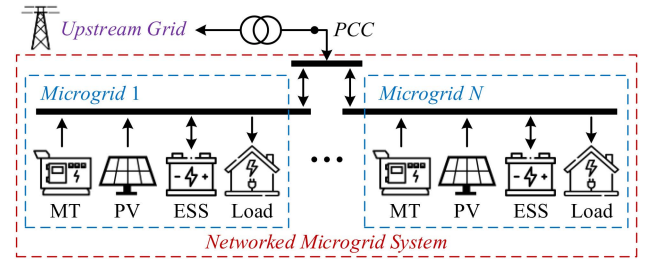


Fig. 1. Illustration of a typical NMG system.

In real-time, the actual reactive power is determined by a local Q/V droop-based control strategy.

- 3) For offline training, both stages are formulated as a multi-agent DRL (MADRL) framework using the soft actor-critic (SAC) algorithm. To ensure policy’s safety, a safety model is developed to support the learning algorithm. For online application, each trained agent works for an individual microgrid in a decentralized manner, without the need for information exchange with other microgrids. Besides, the trained safety model is also able to monitor and update actions during the online application.

The rest of this paper is organized as follows. Section II formulates the studied coordination problem in the NMG. Section III introduces the proposed methodology in detail. Case studies are presented for validation in Section IV. Finally, Section V concludes the paper.

II. PROBLEM DESCRIPTION AND FORMULATION

Fig. 1 gives a general illustration of a typical NMG system architecture where several single microgrids are connected to the point of common coupling (PCC) and each single microgrid consists of MTs, ESSs, PVs, and local loads. In this section, active power management (*P*-management) in the NMG is formulated firstly and a P/SOC droop-based ESS control strategy is designed. Afterwards, reactive power management (*Q*-management) problem in the NMG is modelled, where a Q/V droop-based PV inverter control is considered.

A. Coordination of Networked-Microgrids

In this study, the aim of the coordination is to minimize the total operation costs and active power losses in the NMG system. For the *i*th microgrid ($i = 1, 2, \dots, N$), its cost function C_{MGi} is modeled as:

$$C_{MGi} = \sum_{t=1}^T (C_{MTi}^t + C_{ESSi}^t + C_{TRI}^t) \quad (1)$$

$$C_{MTi}^t = a_i (P_{MTi}^t \Delta t)^2 + b_i (P_{MTi}^t \Delta t) + c_i \quad (2)$$

$$C_{ESSi}^t = d_i (P_{ESSi}^t \Delta t)^2 \quad (3)$$

$$C_{TRI}^t = \lambda_B^t P_{INI}^t \Delta t - \lambda_S^t P_{OUTi}^t \Delta t \quad (4)$$

where C_{MTi}^t , C_{ESSi}^t , C_{TRI}^t are the costs of MT, ESS, and energy trading, P_{MTi}^t , P_{ESSi}^t are MT, ESS output powers,

$P'_{\text{IN}i}$, $P'_{\text{OUT}i}$ refer to active powers import/export to/from the microgrid, λ^t_B , λ^t_S indicate external buying, selling prices in the NMG system, and a_i , b_i , c_i , d_i are positive cost coefficients. Besides, the following operational constraints are considered:

$$0 \leq P_{\text{MT}i}^t \leq P_{\text{MT}i}^{\max}, \forall t \quad (5)$$

$$P_{\text{ESS}i}^t = P_{\text{DIS}i}^t - P_{\text{CH}i}^t, \forall t \quad (6)$$

$$0 \leq |P_{\text{DIS}i}^t| \leq P_{\text{ESS}i}^{\max}, 0 \leq |P_{\text{CH}i}^t| \leq P_{\text{ESS}i}^{\max}, \forall t \quad (7)$$

$$E_i^t = E_i^{t-1} - (P_{\text{DIS}i}^t / \eta_{\text{DIS}} - P_{\text{CH}i}^t \cdot \eta_{\text{CH}}) \Delta t, \forall t \quad (8)$$

$$\text{SoC}_i^{\min} \leq \text{SoC}_i^t \leq \text{SoC}_i^{\max}, \text{SoC}_i^t = E_i^t / E_i^{\text{rate}}, \forall t \quad (9)$$

$$0 \leq |P_{\text{IN}i}^t| \leq P_{\text{TR}i}^{\max}, 0 \leq |P_{\text{OUT}i}^t| \leq P_{\text{TR}i}^{\max}, \forall t \quad (10)$$

where $P_{\text{MT}i}^{\max}$, $P_{\text{ESS}i}^{\max}$ are the maximum power limits of MT, ESS, $P_{\text{TR}i}^{\max}$ is the maximum power exchange value for energy trading, η_{DIS} , η_{CH} are discharging, charging efficiency, E_i^t is the capacity of ESS, and E_i^{rate} refers to the rated capacity.

Due to high-level penetration of RESs in the NMG, voltage regulation as well as active power losses are inevitable. This paper also aims to minimize the network power loss in the system, while satisfying operational limits, which can be modeled as follows:

$$\min \sum_t P_L^t \quad (11)$$

subject to:

$$P_{j+1} = P_j - R_j \frac{P_j^2 + Q_j^2}{V_j^2} - p_{j+1} \quad (12)$$

$$Q_{j+1} = Q_j - X_j \frac{P_j^2 + Q_j^2}{V_j^2} - q_{j+1} \quad (13)$$

$$V_{j+1}^2 = V_j^2 - 2(R_j P_j + X_j Q_j) + (R_j^2 + X_j^2) \frac{P_j^2 + Q_j^2}{V_j^2} \quad (14)$$

$$p_{j+1} = p_{j+1}^l - p_{j+1}^g, q_{j+1} = q_{j+1}^l - q_{j+1}^g \quad (15)$$

$$p_{j+1}^t = p_{j+1}^{l,t} - p_{j+1}^{g,t} \quad (16)$$

$$q_{j+1}^t = q_{j+1}^{l,t} - q_{j+1}^{g,t} \quad (17)$$

$$P_L^t = \sum_j R_j \frac{P_j^2 + Q_j^2}{V_j^2} \quad (18)$$

$$q_{j+1}^{g,t} \approx Q_{j+1}^{t,\text{PV}}, |Q_{j+1}^{t,\text{PV}}| \leq \sqrt{S_{j+1}^2 - P_{j+1}^{t,\text{PV}}^2} \quad (19)$$

$$V_j^{\min} \leq V_j^t \leq V_j^{\max} \quad (20)$$

where P_L^t is the power loss during t -th time interval, π refers to the policy of the agent, P_j , Q_j are active, reactive powers from node j to $j+1$, V_j is the voltage magnitude at node j , R_j , X_j refer to line resistance, reactance between node j and $j+1$, p_j^l , q_j^l are active, reactive power loads at node j , p_j^g , q_j^g are active, reactive power generations at node j , V_S is the primary voltage, $p_j^{l,t}$, $p_j^{g,t}$ are active power load, generation at node j , $q_j^{l,t}$, $q_j^{g,t}$ are reactive power load, generation at node

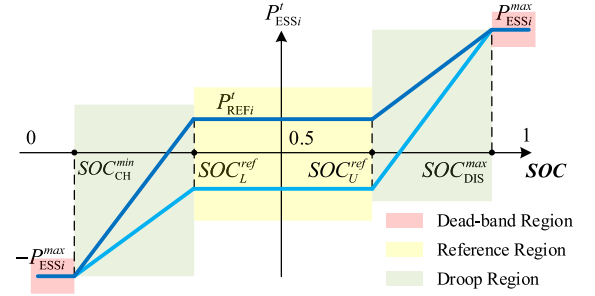


Fig. 2. P/SoC droop-based ESS control.

j , $P'_{j,\text{PV}}$, $Q'_{j,\text{PV}}$ are active, reactive power outputs from PV inverter at node j , $S_{j,\text{PV}}$ is the apparent power size of PV inverter at node j . Equations (12)–(18) describe power flow relationship based on the DistFlow model. Equation (19) represents that the available reactive power capacity of a PV inverter is determined by its apparent power capacity and current active power output. Equation (20) denotes voltage limitations of each node.

B. P/SoC Droop-based ESS Control

A P/SoC droop-based control strategy is applied to balance the SoC level and operation objectives of ESS [31]. Its principle is to adjust the actual output power of ESS according to its SoC. The primary purpose of this droop-based control is to restore SoC under extremely high/low conditions. After adjusting SoC to a normal condition, the output of ESS can be utilized for economic dispatch. This dynamic relationship is described by a piece-wise curve as shown in Fig. 2. The logic-based function is given to illustrate this ESS control as follows:

$$P_{\text{ESS}i}^t = \begin{cases} -P_{\text{ESS}i}^{\max}, & \text{SoC} < \text{SoC}_{\text{CH}}^{\min} \\ P_{\text{REF}i}^t - K_{1,i} (\text{SoC}_L^{\text{ref}} - \text{SoC}), & \text{SoC}_{\text{CH}}^{\min} \leq \text{SoC} < \text{SoC}_L^{\text{ref}} \\ P_{\text{REF}i}^t, & \text{SoC}_L^{\text{ref}} \leq \text{SoC} \leq \text{SoC}_U^{\text{ref}} \\ P_{\text{REF}i}^t + K_{2,i} (\text{SoC} - \text{SoC}_U^{\text{ref}}), & \text{SoC}_U^{\text{ref}} < \text{SoC} \leq \text{SoC}_{\text{DIS}}^{\max} \\ P_{\text{ESS}i}^{\max}, & \text{SoC} > \text{SoC}_{\text{DIS}}^{\max} \end{cases} \quad (21)$$

where $P_{\text{REF}i}^t$ is the reference power of ESS, $\text{SoC}_L^{\text{ref}}$, $\text{SoC}_U^{\text{ref}}$ are lower, upper bounds of SoC to generate provide reference power, $\text{SoC}_{\text{CH}}^{\min}$, $\text{SoC}_{\text{DIS}}^{\max}$ are minimum, maximum values of SoC reaching power limitations, $K_{1,i}$, $K_{2,i}$ indicate droop slopes for different control regions. As plotted in Fig. 2, if SoC is lower than $\text{SoC}_{\text{CH}}^{\min}$ (dead-band region) ESS is managed to restore SoC by charging ESS at the maximum power $-P_{\text{ESS}i}^{\max}$. On the contrary, if SoC is larger than $\text{SoC}_{\text{DIS}}^{\max}$, ESS prefers to support the economic operation at the maximum discharging power $P_{\text{ESS}i}^{\max}$. When SoC is located within the reference region $[\text{SoC}_L^{\text{ref}}, \text{SoC}_U^{\text{ref}}]$, the output power of ESS follows the reference power calculated by the power management agent. If SoC is located within the droop region $[\text{SoC}_U^{\text{ref}}, \text{SoC}_{\text{DIS}}^{\max}]$, the output power of ESS follows a

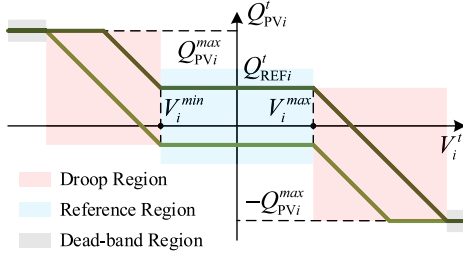


Fig. 3. Q/V droop-based ESS control.

droop-based curve, which can be calculated according to SoC value, reference power, and droop slope. In this paper, SoC_{L}^{ref} , SoC_{U}^{ref} are defined empirically as 30% and 70% respectively. Moreover, the droop slopes $K_{1,i}$, $K_{2,i}$ can be calculated based on determined points $(SoC_{DIS}^{max}, P_{ESSi}^{max})$ ($SoC_{U}^{ref}, P_{REFi}^t$) and $(SoC_{CH}^{min}, -P_{ESSi}^{max})$ ($SoC_{L}^{ref}, P_{REFi}^t$). This ESS control mainly aims to ensure the safety operation by regulating SoC in the rated range (5%, 95%). If SoC is located in the middle range, the output of ESS will follow the reference value, which is optimized for economic dispatch. On the other hand, ESS with lower SoC can restore its capacity promptly; ESS with higher SoC is encouraged to generate more powers into the system. The reference power of ESS P_{REFi}^t is optimized to minimize the generation cost of each individual microgrid as well as total network power loss with a 1-hour resolution. To specify, the proposed method focuses on searching for an optimal ESS reference power P_{REFi}^t as well as MT output P_{MTi}^t .

C. Q/V Droop-Based PV Inverter Control

Due to flexible implementation and fast response speed, PV inverters have been suggested for VVC by the IEEE Std 1547 working group [16]. To allow PV inverters to provide voltage control support, a Q/V droop-based controller is applied in this paper, which aims to achieve a balance between real-time voltage regulation and power loss minimization. Similar as the ESS control, the primary objective of this droop-based control is to restore the voltage if the voltage exceeds the limits. When the voltage is within the normal range, the reactive power is managed mainly according to the reference value, in order to minimize the network power loss. Fig. 3 gives an illustration of this Q/V droop-based PV inverter control, and the logic-based function of the inverter control is given as follows:

$$Q_{PVi}^t = \begin{cases} \min(Q_{REFi}^t + K_{V,i}(V_i^{min} - V_i^t), Q_{PVi}^{max}), & V_i^t < V_i^{min} \\ Q_{REFi}^t, & V_i^{min} \leq V_i^t \leq V_i^{max} \\ \max(Q_{REFi}^t - K_{V,i}(V_i^t - V_i^{max}), -Q_{PVi}^{max}), & V_i^t > V_i^{max} \end{cases} \quad (22)$$

where Q_{PVi}^t is the actual reactive power output of PV inverter in i th microgrid, Q_{PVi}^{max} refers to the maximum reactive power output of PV inverter in i th microgrid, Q_{REFi}^t is the reference reactive power of PV inverter, $K_{V,i}$ is the control gain of the

droop curve. To obtain the droop control gain, the relationship between power injection increment and voltage variations is expressed based on Jacobian matrix as follows [14]:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} K & L \\ M & N \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix} \quad (23)$$

According to (23), the voltage variation can be derived as:

$$\Delta V = (N - MK^{-1}L)^{-1} \Delta Q - (N - MK^{-1}L)^{-1} MK^{-1} \Delta P \quad (24)$$

Therefore, the sensitivity matrix of reactive power injection to the voltage variation can be obtained as:

$$S = \frac{\Delta V}{\Delta Q} = (N - MK^{-1}L)^{-1} \quad (25)$$

The diagonal elements of S indicate the sensitivity of reactive power injection to bus voltage variation. Based on that, the droop gain $K_{V,j}$ (j indicates the node that PV is installed) can be achieved as $K_{V,j} = \omega/s_{jj}$, where ω is a coefficient and can be defined as $1/N_{bus}$. In this paper, the reactive power setpoint of PV inverter is optimized to minimize the active power loss as well as satisfy the operational constraints with each 3-minute interval. Further, the real-time droop-based Q -control is implemented by a local controller inside each PV inverter to regulate the voltage within every 3 minutes.

III. PROPOSED DATA-DRIVEN METHOD

In this chapter, a data-driven coordination methodology is proposed for the NMG system. Firstly, the problem is modelled as Markov Decision Process (MDP). Then, a general framework is briefly introduced. Afterwards, DRL algorithm is introduced to learn the optimal policy of the coordination in NMG. Further, a safety model is developed to avoid voltage violations with a fast time-scale.

A. Markov Decision Process (MDP) Modeling

In this study, each individual microgrid has two types of agents for P/Q management respectively π_P , π_Q . These agents can generate actions based on received states as $a_P = \pi_P(s_P|\theta_P)$, $a_Q = \pi_Q(s_Q|\theta_Q)$, where θ_P , θ_Q are parameters of policy π_P , π_Q . It should be noted that two types of agents are executed in different time-scales, that π_P with a slower time-scale (1-hour) and π_Q with a faster time-scale (3-minute). After new actions are executed in the environment, a reward r will be observed to evaluate the system performance. In order to search for the optimal policy π_P , π_Q , the overall problem is formulated as a Markov Decision Process (MDP) as follows:

1) *Reactive Power Management: State*: To handle the stochastic characteristics of loads and RESs, historic load and PV data are also included in the state construction. Reactive power dispatch is executed in a faster time-scale (3-minute) due to the faster response speed of PV inverters, and the state includes voltage deviation, historic data of active power mismatch,

historic data of reactive powers as:

$$s_{Qi}^t = [\Delta V_i^t, \Delta p_{h,i}^{t-1}, \dots, \Delta p_{h,i}^{t-t_D}, Q_{LOADi}^{t-1}, \dots, Q_{LOADi}^{t-t_D}] \quad (26)$$

$$\Delta V_i^t = V_i^t - V_{REF} \quad (27)$$

$$\Delta p_{h,i}^{t-t_D} = P_{MTi}^{t-t_D} + P_{ESSi}^{t-t_D} + P_{PVi}^{t-t_D} - P_{LOADi}^{t-t_D} \quad (28)$$

where t indicates the time-scale for Q -management (3-minute), ΔV_i^t refers to the voltage deviation, V_{REF} indicates the nominal voltage, $\Delta p_{h,i}^{t-t_D}$ denotes historic data of active power mismatch, Q_{LOADi}^t denotes historic data of reactive load demand, and t_D denotes the length of delayed time step for historic data. Besides, P_{MT}^t , P_{ESS}^t are determined by P -management agents.

Action: The purpose of Q -agent is to manage the reactive power output of PV inverters. Based on the droop-based inverter control, the designed action is defined as the reference reactive power of the droop-based curve:

$$a_{Qi}^t = Q_{REFi}^t \quad (29)$$

The time-resolution of the action a_Q is defined as 3 minutes. Normally, the dispatch interval of PV inverters is set as 15 minutes [14], [16] for optimization-based methods. Since the data-driven method is applied in this paper that only needs matrix calculation at online stage, it can greatly release the online computational burden. Therefore, a relatively short timescale (3-minute) is selected in this stage. In addition, the PV-inverter management actually consists of two stages: 3-minute Q -management by the data-driven agent and real-time Q/V droop-based controller given in (22).

Reward: To evaluate the performance of an action-state pair (s_Q, a_Q) and guide the policy learning, a global reward is defined based on (18) as follows:

$$r_q^t = -\alpha_1 P_L^t \quad (30)$$

where α_1 is the coefficient of the reward. Based on this reward, all agents are trained towards minimizing the overall active power loss during offline learning.

2) **Active Power Management:** **State:** Active power dispatch is executed in a slower time-scale (1-hour). To consider the stochastic characteristics of loads and RESs, historic load and PV data are also included in the state construction as follows:

$$s_{Pi}^T = [\lambda_B^T, \lambda_S^T, SoC_i^T, \Delta p_i^{T-1}, \Delta p_i^{T-2}, \dots, \Delta p_i^{T-T_D}] \quad (31)$$

$$\Delta p_i^{T-T_D} = P_{PVi}^{T-T_D} - P_{LOADi}^{T-T_D} \quad (32)$$

where T is the time-scale of P management (1-hour), two items in (32) are past data of PV generation and active load demand respectively. $\Delta p_i^{T-T_D}$ indicates the mismatch between PV generation and load demand, T_D denotes the length of delayed time step for historic data. By adding historic information into the state, the agent implicitly integrates the function of PV and load forecasting.

Action: The P -agent aims to manage the active power outputs of MTs and ESSs. In this regard, the action is defined based on the setpoint of MT and the reference power of the droop-based

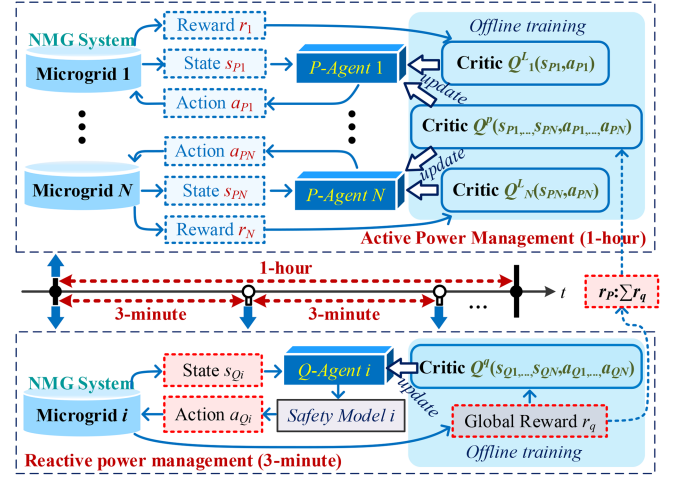


Fig. 4. Framework of the proposed data-driven method.

ESS control curve:

$$a_{Pi}^T = [P_{MTi}^T, P_{REFi}^T] \quad (33)$$

The time-resolution of the action a_P is defined as 1 hour. In addition, ESS management consists of two stages: 1-hour P -management by the data-driven agent and real-time P/SoC droop-based controller given in (21).

Reward: To give the right direction of policy π_P learning, two types of reward, a local reward and a global reward, are defined for P -agent, as follows:

$$r_i^T = -\alpha_2 C_{MGi} \quad (34)$$

$$r_p^T = -\alpha_3 \lambda_B^T \sum_t r_q^t \quad (35)$$

where r_i^T denotes the local reward of i th microgrid, r_p^T denotes the global reward after the environment receives all actions, and α_2, α_3 are coefficients of rewards. Based on local rewards r_i^T , the optimal policy of each P -agent can be learned towards minimizing the local generation cost. Meanwhile, through the global reward r_p^T , each agent can also coordinate to minimize the network loss. Furthermore, both P - and Q -management should satisfy operational constraints, i.e. (5)–(10) (12)–(20).

B. General Framework

Fig. 4 gives a general framework of the proposed data-driven method. The active power outputs of MTs and ESSs are managed in a slow time-scale (1-hour) and reactive power outputs of PV inverters are dispatched in a fast time-scale (3-minute). Hence, two types of agents are trained for P and Q management respectively.

For P -management, each agent is initialized based on a Deep Neural Network (DNN). After receiving the state from the environment (NMG system), the agent will generate the action and send it back to the environment for execution. The actions of P -management agents are fixed in one hour. At a faster time-scale, Q -management agent receives the outputs from the

upper level and local observed states to generate reactive power setpoints. Different from P -management, the action can be monitored and modified by a safety model to avoid voltage violations. By executing the actions for Q -management, a global reward can be observed to evaluate the performance. By accumulating this global reward, another global reward can be achieved to update P -management agents, as well as local rewards. After certain number of iterations, transitions (i. e., state, action, reward) are memorized into replay buffers. Based on replay buffers, agents can be updated by maximizing accumulated rewards through policy learning.

After numerical learning iterations, optimal policies can be learned for agents, and these trained agents can be applied for online application, where P - and Q -agents operate in different time-scales. Both P - and Q -agents can work in a decentralized manner at online stage, that each agent only utilizes local observation as the inputs and generates the action for each corresponding microgrid. Further, the trained safety model can also be applied for safety monitoring and update the agents at online stage. Besides, P and Q controls would not conflict under this hierarchical structure. In this paper, active and reactive power are dispatched within different timescales. Also, the devices for active and reactive power management are also different: ESSs, MTs for P -management and PV inverters for Q -management.

C. Policy Learning By Multi-Agent Soft Actor-Critic

In this study, RL algorithm is applied to conduct the policy learning. For most RL algorithms, the actor-critic structure has been extensively utilized, consisting of an actor network and a critic network. As well-known actor-critic-based algorithms, deep deterministic policy gradient (DDPG), asynchronous advantage actor-critic (A3C), soft actor-critic (SAC) have been studied and applied in numerous research works. Unlike other algorithms, SAC has a better exploration capability as it aims to maximize the entropy of the policy as well as the accumulate reward, Which can be expressed as:

$$\pi^* = \arg \max_{\pi} \sum E_{(s,a)} [r + \tau \cdot H(\pi(s|\theta))] \quad (36)$$

where τ is a hyperparameter indicating the weight of the entropy, and $H(\cdot)$ is the function of entropy as:

$$H(\pi(s|\theta)) = E_{a \sim \pi(s|\theta)} [-\log \pi(s|\theta)] \quad (37)$$

To evaluate the performance of a pair of state and action (s,a) , a critic network is normally built to approximate the state-action value function as $Q(s,a|\phi)$, where ϕ is the parameters of this critic network. The target of this critic network $Q_{Tar}(s,a)$ is shown as:

$$Q_{Tar}(s,a) = E_{s,a,r \in \mathbf{R}} \left[\sum \gamma \cdot r + \sum \gamma \cdot H(\pi(s|\theta)) \right] \quad (38)$$

where $\mathbf{R}$ is the replay buffer collected from actor-environment interactions, and γ indicates the discount factor. Normally, an entropy-regularized Bellman equation is derived as the target function of the critic network. Besides, three types of critic networks (Q^L , Q^p , Q^q) are developed, which indicate the local benefit of P -management, social benefit of P management, and social benefit of Q management, as follows:

Based on Bellman equation shown in (39)–(41) shown at the bottom of the this page, these critic network can be trained according to the loss function as:

$$\begin{aligned} J_{Q^L}(\phi_i^L) &= E_{s_{Pi}^T, a_{Pi}^T \in \mathbf{R}} \left[\frac{1}{2} (Q_i^L(s_{Pi}^T, a_{Pi}^T | \phi_i^L) - Q_{Tar,i}^L(s_{Pi}^T, a_{Pi}^T))^2 \right] \end{aligned} \quad (42)$$

$$\begin{aligned} J_{Q^p}(\phi^p) &= E_{s_{Pi}^T, a_{Pi}^T \in \mathbf{R}} \left[\frac{1}{2} \left(Q^p(s_{P1}^T, \dots, s_{Pi}^T, a_{P1}^T, \dots, a_{Pi}^T | \phi^p) - Q_{Tar}^p(s_{P1}^T, \dots, s_{Pi}^T, a_{P1}^T, \dots, a_{Pi}^T) \right)^2 \right] \end{aligned} \quad (43)$$

$$\begin{aligned} J_{Q^q}(\phi^q) &= E_{s_{Qi}^t, a_{Qi}^t \in \mathbf{R}} \left[\frac{1}{2} \left(Q^q(s_{Q1}^t, \dots, s_{Qi}^t, a_{Q1}^t, \dots, a_{Qi}^t | \phi^q) - Q_{Tar}^q(s_{Q1}^t, \dots, s_{Qi}^t, a_{Q1}^t, \dots, a_{Qi}^t) \right)^2 \right] \end{aligned} \quad (44)$$

$$Q_{Tar,i}^L(s_{Pi}, a_{Pi}) = E_{\substack{s_{Pi}^{T+1}, r_i^T \in \mathbf{R} \\ a_{Pi}^{T+1} = \pi(s_{Pi}^{T+1} | \theta_{Pi})}} \left\{ r_i^T + \gamma \left[Q_i^L(s_{Pi}^{T+1}, a_{Pi}^{T+1}) + \tau \cdot H(\pi(s_{Pi}^{T+1} | \theta_{Pi})) \right] \right\} \quad (39)$$

$$\begin{aligned} Q_{Tar}^p(s_{P1}, \dots, s_{Pi}, a_{P1}, \dots, a_{Pi}) &= E_{\substack{s_{Pi}^{T+1}, r_p^T \in \mathbf{R} \\ a_{Pi}^{T+1} = \pi_{Pi}(s_{Pi}^{T+1})}} \left\{ r_p^T + \gamma \left[Q^p(s_{Pi}^{T+1}, \dots, s_{Pi}^{T+1}, a_{Pi}^{T+1}, \dots, a_{Pi}^{T+1}) + \tau \cdot \sum_{i=1}^N H(\pi_{Pi}(s_{Pi}^{T+1} | \theta_{Pi})) / N \right] \right\} \end{aligned} \quad (40)$$

$$\begin{aligned} Q_{Tar}^q(s_{Q1}, \dots, s_{Qi}, a_{Q1}, \dots, a_{Qi}) &= E_{\substack{s_{Qi}^{t+1}, r_q^t \in \mathbf{R} \\ a_{Qi}^{t+1} = \pi_{Qi}(s_{Qi}^{t+1})}} \left\{ r_q^t + \gamma \left[Q^q(s_{Qi}^{t+1}, \dots, s_{Qi}^{t+1}, a_{Qi}^{t+1}, \dots, a_{Qi}^{t+1}) + \tau \cdot \sum_{i=1}^N H(\pi_{Qi}(s_{Qi}^{t+1} | \theta_{Qi})) / N \right] \right\} \end{aligned} \quad (41)$$

Furthermore, two types of actor networks are initialized for P - and Q -management respectively. The actor networks are designed as probability distributions as:

$$a_{P_i}^T = \pi_{P_i}(s_{P_i}^T, \xi_{P_i}^T) = \tanh(\mu_{P_i}(s_{P_i}^T) + \sigma_{P_i}(s_{P_i}^T) \cdot \xi) \quad (45)$$

$$a_{Q_i}^t = \pi_{Q_i}(s_{Q_i}^t, \xi_{Q_i}^t) = \tanh(\mu_{Q_i}(s_{Q_i}^t) + \sigma_{Q_i}(s_{Q_i}^t) \cdot \xi) \quad (46)$$

where μ, σ indicate parameterized DNNs to generate mean and covariance, ξ is a random variable sampled from a standard normal distribution, and $\tanh(\cdot)$ denote the hyperbolic tangent function. By constructing the actor networks based on normal distribution, the policy can be well explored in a stochastic manner. Noted that both state and action spaces are implemented in the continuous space. To realize the policy learning, the actor networks are updated based on following loss functions:

$$\begin{aligned} J_{\pi_{P_i}}(\theta_{P_i}) &= \mathbb{E}_{s_{P_i}^T, a_{P_i}^T \in \mathbf{R}} \left[\log \pi_{P_i}(s_{P_i}^T | \theta_{P_i}) - Q_i^L(s_{P_i}^T, \pi_{P_i}(s_{P_i}^T | \theta_{P_i})) \right] \\ &\quad - Q^p(s_{P_1}^T, \dots, s_{P_i}^T, a_{P_1}^T, \dots, \pi_{P_i}(s_{P_i}^T | \theta_{P_i})) \end{aligned} \quad (47)$$

$$\begin{aligned} J_{\pi_{Q_i}}(\theta_{Q_i}) &= \mathbb{E}_{s_{Q_i}^t, a_{Q_i}^t \in \mathbf{R}} \left[\log \pi_{Q_i}(s_{Q_i}^t | \theta_{Q_i}) \right. \\ &\quad \left. - Q^q(s_{Q_1}^t, \dots, s_{Q_i}^t, a_{Q_1}^t, \dots, \pi_{Q_i}(s_{Q_i}^t | \theta_{Q_i})) \right] \end{aligned} \quad (48)$$

As shown in (47), the agent managing the active power π_{P_i} is updated by maximizing a local objective along with a global objective. Similar, the Q -management agent is also trained in a multi-agent manner based on a global Q -value function. As a maximum entropy algorithm, SAC can seek a wider action space compared with deterministic algorithms, which can achieve a better exploration for offline training.

D. Safety Model

As discussed, SAC can perform a wide action exploration by policy's entropy maximization. However, the exploration process could lead to unsafe actions, which violate operational constraints. This unsafety could lead to high learning expense at offline stage and infeasible solutions at online stage. Hence, to achieve a trade-off between policy's exploration and action's safety is a critical issue for RL algorithms. In existing works, the penalty-based reward or a Lagrange optimization function is applied to avoid constraint violations, focusing on the improvement of offline learning [32], [33]. However, these approaches cannot monitor or update online actions, especially under unlearned scenarios. Besides, some publications also developed the safety model for online application [29], [30], which may require the prior knowledge for assistance.

To solve the mentioned issue, a safety model is developed in this paper, which is designed based on a fully data-driven approach and operates as an intermediate layer between action's generation and execution. It should be noted that the safety model is applied for Q -management agent. Besides, voltage

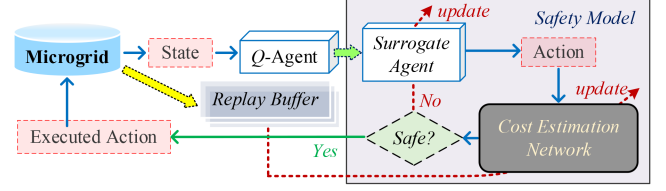


Fig. 5. The overlook of the proposed safety model.

constraint is the main concentration in this study while other constraints have been considered in the environment/ agent setting. Fig. 5 gives a general overlook of the proposed safety model. As the basic component part of this safety model, a constraint estimation network $\Phi(s_{Q_i}^t, a_{Q_i}^t | \Omega)$ is built to predict the condition of voltage violation based on a pair of state and action $(s_{Q_i}^t, a_{Q_i}^t)$, which can be expressed as:

$$\Phi_{Tar}(s_{Q_i}^t, a_{Q_i}^t | \Omega) = (V^{t+1} - V_{REF})^2 \quad (49)$$

where Ω indicates parameters of the constraint estimation network. Based on historic data from the replay buffer, the training of the network is relied on the loss function as following:

$$J_{\Phi}(\Omega) = \mathbb{E}_{s_{Q_i}^t, a_{Q_i}^t \in \mathbf{R}} \left[\frac{1}{2} (\Phi(s_{Q_i}^t, a_{Q_i}^t | \Omega) - \Phi_{Tar})^2 \right] \quad (50)$$

Once the constraint estimation network is trained, the cost value can be predicted before the action is executed. If the predicted value is lower than the threshold $\Phi(s_{Q_i}^t, a_{Q_i}^t | \Omega) \leq \Phi_{th}$, the action is considered as a safe action and can be executed in the environment. If the predicted value exceeds the threshold Φ_{th} , the action is regarded as unsafe. Under this circumstance, a surrogate agent is generated by copying an original Q -agent, and this surrogate agent is adjusted by minimizing the cost function value Φ as:

$$J_S(\theta_{Q_i}) = \mathbb{E}_{s_{Q_i} \in \mathbf{R}} [\Phi_i(s_{Q_i}, \pi_{Q_i}(s_{Q_i} | \theta_{Q_i}) | \Omega_i)] \quad (51)$$

After updating the agent, a new action will be generated and sent back to the constraint estimation network. This iterative process will be continued until a safe action is achieved, which can be utilized as an executed action in the environment. It should be noted that the original Q -agent will not be affected in this process.

The constraint estimation network Φ is trained along with policy learning by SAC. As the learning episode increases, the replay buffer becomes larger by collecting more datasets, which could lead to a thorough training of the network Φ . For offline training, this safety model can satisfy the operational constraints during policy exploration and hence accelerate the learning process. Therefore, the trained network can also be applied at online stage to monitor and improve online actions for safety compliance.

E. Training Process

The offline training of the proposed algorithm is realized in a multi-agent framework. To initialize, two types of actor

networks are constructed as $a^T_{Pi} = \pi_{Pi}(s^T_{Pi})$, $a^t_{Qi} = \pi_{Qi}(s^t_{Qi})$, and three types of critic networks are also constructed based on (39)–(41) as $Q^L_i(s^T_{Pi}, a^T_{Pi})$, $Q^p(s^T_{P1}, \dots, s^T_{PN}, a^T_{P1}, \dots, a^T_{PN})$, $Q^q(s^t_{Q1}, \dots, s^t_{QN}, a^t_{Q1}, \dots, a^t_{QN})$. Besides, the safety model is initialized as a DNN, $\Phi_i(s^t_{Qi}, a^t_{Qi})$. At the beginning of each learning episode, random initial states $\{s^T=1_{P1}, s^T=1_{P2}, \dots, s^T=1_{PN}\}$ are selected for individual microgrids. P -management agents generate and execute all actions $\{a^T_{P1}, a^T_{P2}, \dots, a^T_{PN}\}$ based on observed states every 1 hour. Within 1 hour, Q -management agents generate actions $\{a^t_{Q1}, a^t_{Q2}, \dots, a^t_{QN}\}$ every 3 minutes. Before executed, these actions will be sent to constraint estimation networks of safety models. If the estimated cost is lower than the threshold Φ_{th} , the action can be executed directly and next states s^{t+1}_{Qi} will be observed. If the estimated cost is beyond the threshold Φ_{th} , the actor network shall be updated based on (51) to achieve a safe policy. After actions are executed, a global reward r^t_q can be achieved to minimize the total power network loss based on (30). At the end of 1 hour, another global reward r^T_p is achieved by accumulating r^t_q and local rewards r^T_i are calculated based on (34) for P -management agents. Besides, new states s^{T+1}_{Pi} can be observed for the next hour (iteration). To store those transitions, two types of replay buffers are built for each microgrid as $\{s^T_{Pi}, a^T_{Pi}, r^T_p, s^{T+1}_{Pi}\}$, $\{s^t_{Qi}, a^t_{Qi}, r^t_q, s^{t+1}_{Qi}\}$ ($i = 1, 2, \dots, N$). According to replay buffers, critic networks are trained to approximate state-action functions by (42)–(44). Then, actor networks are updated by maximizing state-actions values based on (47), (48). Furthermore, constraint estimation networks are also trained in each iteration according to the loss function (50).

The overall proposed algorithm is shown in Algorithm 1 as follows:

IV. CASE STUDY

In this chapter, case studies are carried on to validate the performance of the proposed method, which are performed on a laptop with AMD Ryzen 7 4800H 2.9 GHz CPU and 16 GB RAM. Both offline training and testing results are presented.

A. Simulation Setup

Based on IEEE 33-bus radial distribution network, a NMG system is built by four single microgrids, as plotted in Fig. 6. Each microgrid has a MT, an ESS, a PV, and a local load. Table I gives main parameters of the studied system. The nonlinear AC power flow model of the system is simulated by MATPOWER [34].

For each microgrid, PV and load data are injected, which are extracted from 1-year historic database. Fig. 7 illustrates an example of 1-week load, PV data in microgrid 1. In each episode, 3-day PV/load data are randomly extracted from the database and injected into the test system. Note that input PV/load data for training and testing are not overlapped. Further, external buying/selling prices of 24-hour are also presented in Fig. 8.

Algorithm 1: Coordination Algorithm Based on Safe DRL.

1: Initialize:

Active power management:

Agents: $\pi_{Pi} \sim \theta_{Pi}$, $i = 1, 2, \dots, N$

Critics: $Q^L_i \sim \phi^L_i$ (local benefit), $i = 1, 2, \dots, N$

$Q^p \sim \phi^p$ (global benefit)

Reactive power management:

Agents: $\pi_{Qi} \sim \theta_{Qi}$, $i = 1, 2, \dots, N$

Critics: $Q^q \sim \phi^q$ (global benefit)

Safety model: $\Phi_i \sim \Omega_i$, $i = 1, 2, \dots, N$

Replay buffer: $\mathbf{R}_p, \mathbf{R}_q$

2: for episode = 1 to N_{epi} do:

3: Randomly select initial states $\{s^1_i\}$

4: for $T = 1$ to 24 (every hour) do:

5: Generate and execute actions for P-management

$a^T_{Pi} = \pi_{Pi}(s^T_{Pi}|\theta_{Pi})$

6: for $t = 1$ to 20 (every 3 minute) do:

7: Generate actions for Q-management

$a^t_{Qi} = \pi_{Qi}(s^t_{Qi}|\theta_{Qi})$

8: Predict the cost $c^t_i = \Phi_i(s^t_{Qi}, a^t_{Qi})$

9: if $c^t_i > \Phi_{th}$ then:

10: Update Q-management agents with learning

rate λ_ϕ : $\theta_{Qi} \leftarrow \theta_{Qi} - \lambda_\phi \nabla_{\theta_{Qi}} J_S(\theta_{Qi})$

11: Return to 7

12: end if

13: Execute action a^t_{Qi} , get a global reward r^t_q , observe new states s^{t+1}_{Qi}

14: Store transition $\{s^t_{Qi}, a^t_{Qi}, r^t_q, s^{t+1}_{Qi}\}$ into $\mathbf{R}_q$

15: $t \leftarrow t + 1$

16: end for

17: Observe a global reward r^T_p , local rewards r^T_i , and new states s^{T+1}_{Pi}

18: Store transition $\{s^T_{Pi}, a^T_{Pi}, r^T_p, s^{T+1}_{Pi}\}$ into $\mathbf{R}_p$

19: $T \leftarrow T + 1$

20: end for

21: Mini-batches m_p, m_q are sampled from $\mathbf{R}_p, \mathbf{R}_q$

22: Update critic networks with learning rate λ_Q :

$\phi^L_i \leftarrow \phi^L_i - \lambda_Q \nabla_{\phi^L_i} J_{QL_i}(\phi^L_i)$

$\phi^p \leftarrow \phi^p - \lambda_Q \nabla_{\phi^p} J_{Qp}(\phi^p)$

$\phi^q \leftarrow \phi^q - \lambda_Q \nabla_{\phi^q} J_{Qq}(\phi^q)$

23: Update agents with learning rate λ_π :

$\theta_{Pi} \leftarrow \theta_{Pi} - \lambda_\pi \nabla_{\theta_{Pi}} J_{\pi_{Pi}}(\theta_{Pi})$,

$\theta_{Qi} \leftarrow \theta_{Qi} - \lambda_\pi \nabla_{\theta_{Qi}} J_{\pi_{Qi}}(\theta_{Qi})$

24: Update the safety model with learning rate λ_S :

$\Omega_i \leftarrow \Omega_i - \lambda_S \nabla_{\Omega_i} J_{\Phi_i}(\Omega_i)$

25: end for

B. Offline Training

For offline training, several DNNs are initialized, including actor networks, critic networks, constraint estimation networks. Then, these DNNs are sent to MATLAB interface for iterative RL learning process (agent-environment interaction). At the beginning of each episode, the starting day is randomly selected based on historic database. After multiple interactions between agent and environment, transitions are stored into the replay

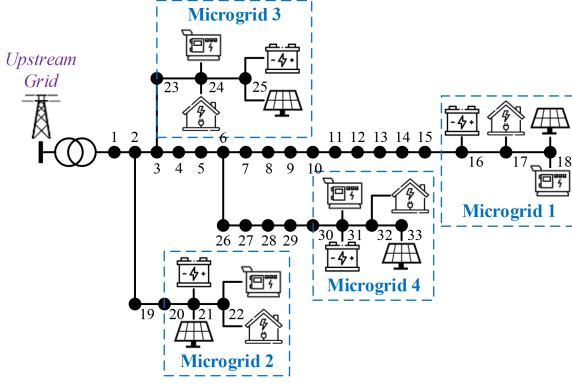


Fig. 6. NMG test model based on IEEE 33-bus distribution network.

TABLE I
MAIN PARAMETERS OF THE TEST MODEL

Parameters	Values
a_i, b_i, c_i	0.0002 \$/kWh ² , 0.03 \$/kWh, 20 \$
d_i	0.005 \$/kWh ²
P_{MTi}^{max}	1 MW
P_{ESSi}^{max}	400 kW
η_{DIS}, η_{CH}	0.97, 0.95
E_i^{rate}	2 MW
$SoC_{min}^{min}, SoC_{max}^{max}$	5 %, 95 %
$SoC_{Li}^{ref}, SoC_{U}^{ref}$	30 %, 70 %
V_{min}, V_{max}	0.95 p.u., 1.05 p.u.

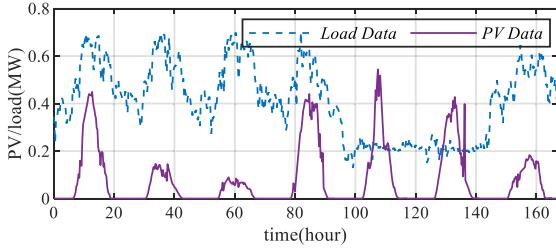


Fig. 7. 1-week input profiles of PV/load historic data in microgrid 1.

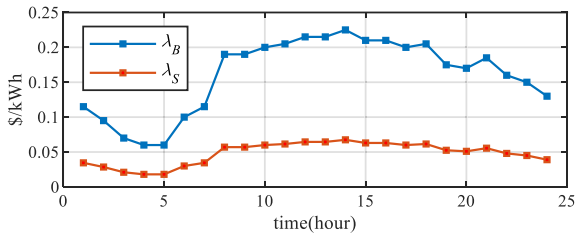


Fig. 8. External buying/selling prices in 24 hours.

buffer. Then, a mini-batch is extracted from this replay buffer and sent to Python interface to implement the policy learning. This Python-based policy learning is programmed and conducted by Pytorch library. The parameters of DRL setting and offline training are provided in Table II. After policy learning, updated DNNs will be sent back to MATLAB interface for the next episode.

TABLE II
PARAMETERS OF DRL SETTING AND OFFLINE TRAINING

Parameters	Value	Parameters	Value
π_p structure	23×512×128×2	π_Q structure	25×512×128×1
Φ structure	26×512×128×1	Q^L structure	25×512×128×1
Q^p structure	104×512×128×1	Q^L structure	100×512×128×1
Learning rate λ_Φ	1e-4	R_p size	2×10 ⁴
Learning rate λ_Q	1e-4	R_q size	5×10 ⁴
Learning rate λ_π	1e-5	m_p size	2×10 ³
Learning rate λ_S	1e-3	m_q size	5×10 ³
Threshold Φ_{th}	0.05	Maximum N_{epi}	5×10 ⁴
Discount factor γ	0.99	Entropy weight τ	0.02

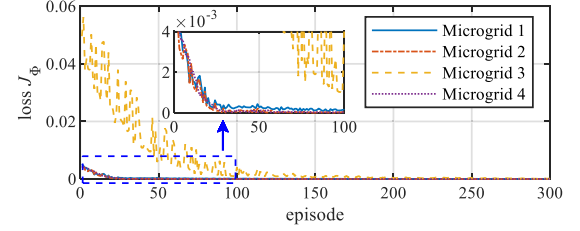


Fig. 9. Loss function values of constraint estimation networks.

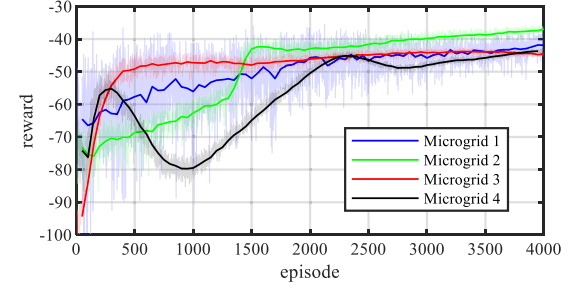


Fig. 10. Reward of local benefit.

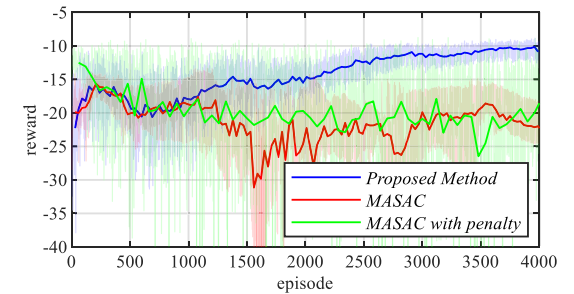


Fig. 11. Reward of global benefit.

Constraint estimation networks are trained simultaneously during policy learning, whose loss function values are shown in Fig. 9. As shown in the figure, the training of constraint estimation networks is about to be converged after 100 episodes. Therefore, the safety model begins to work after 100 episodes since constraint estimation networks can predict the cost accurately according to a pair of state and action. Also, constraint estimation networks will continue updating in following episodes. Fig. 10 presents local rewards during offline training. Besides, Fig. 11

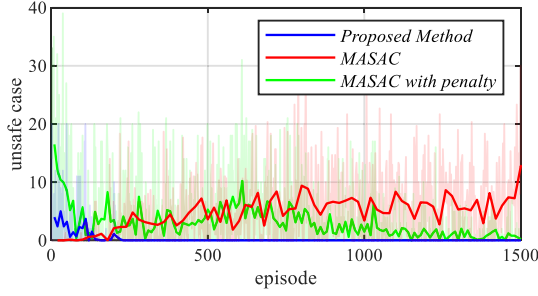


Fig. 12. Number of unsafe cases in offline training.

plots global rewards for social benefits, learned by the proposed method, multi-agent SAC (MASAC), and MASAC with penalty-based reward, where rewards for every episode are plotted by light color and average rewards per 20 episodes are plotted by deep color. As presented in Fig. 11, the proposed method shows an overall stable learning performance and achieves a well convergence. On the other hand, MASAC without safety model led the whole learning process to another learning paths, which could result in high learning cost. Besides, if the penalty item is added into the reward, the algorithm may not perform well with an unstable process. To further verify the functionality of the safety model, Fig. 12 provides the number of unsafe cases during training. Note that if voltage constraints are violated, the whole operation under this 3-minute time-scale is considered as an unsafe case. The proposed method shows a safer exploration process that there is no unsafe case after around 300 episodes. However, MASAC without safety models hardly eliminate these unsafe actions as shown in Fig. 12. According to results above, the proposed algorithm is capable to perform an effective and advantageous learning process. Moreover, different from those optimization-based methods that could be adversely affected by PV/load prediction errors, the proposed method already includes the predictions of PV generations and load demands into the state definition, which can get rid of extra prediction modules. Also, the data-driven method only needs to conduct matrix calculation at online stage after offline training, which has less computational burden than optimization-based methods. Based on offline tests, the average computational time is around 0.01 s level that is suitable for online application. Besides, the proposed method is realized in a decentralized way, which also releases much communication burden.

C. Test Results

To evaluate the performance of the proposed method, several tests are conducted in this section, where input data (PV, load data) do not overlap with the data for training. PV/load data for testing are shown in Fig. 13. The initial capacities of ESSs are set as 500 kWh, 1000 kWh, 1000 kWh, 2000 kWh at t_0 moment, and corresponding SOC values are 25%, 50%, 50%, 75%. By utilizing historic data and system state as inputs, the trained P -management agent will manage the setpoints of MT, ESS in a decentralized manner. Fig. 14 presents P -management results in each microgrid. When PV generation is relatively high

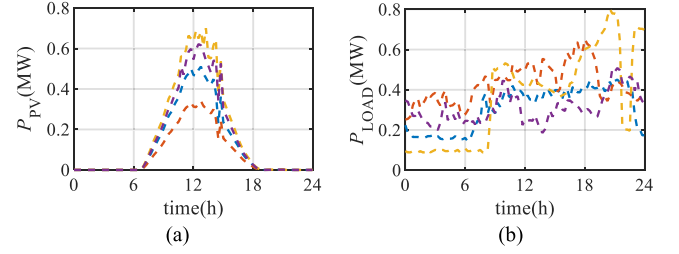


Fig. 13. 1-day input profiles of PV/load historic data in case 1 (a) PV data (b) load data.

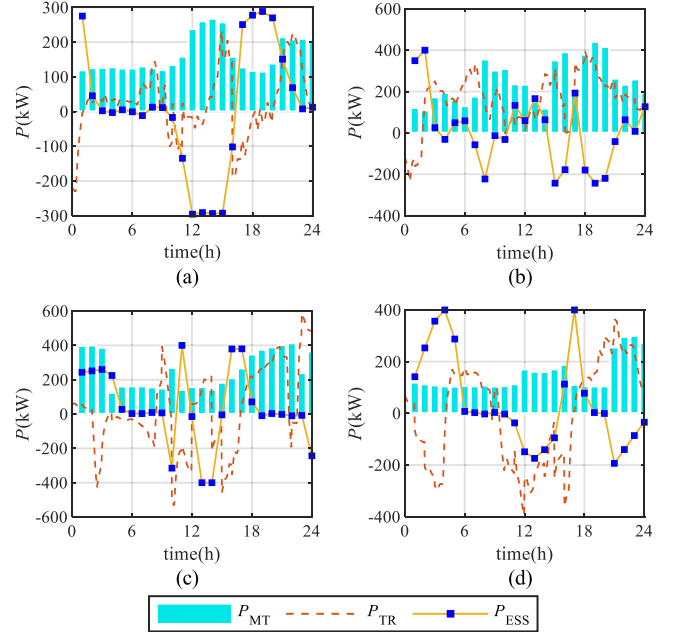


Fig. 14. Active power outputs: (a) microgrid 1, (b) microgrid 2, (c) microgrid 3, and (d) microgrid 4.

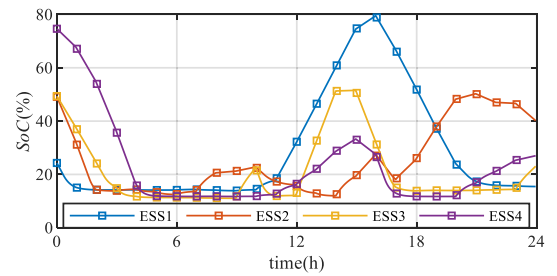


Fig. 15. SoC value of ESS in each individual microgrid.

(around 12:00), ESS prefers to charge and individual microgrid tends to sell surplus power to the main grid. Also, SOC of each ESS is shown in Fig. 15, which satisfies the operational constraints. To validate the performance of droop-based ESS control, Fig. 16 gives an illustration of ESS control curves at the starting point. For example, the reference power of ESS in microgrid 1 is defined as a positive value, indicating ESS is encouraged to discharge to support the system. However,

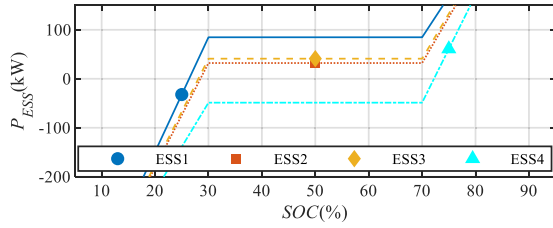


Fig. 16. ESS control curves and corresponding operating points at the beginning moment.

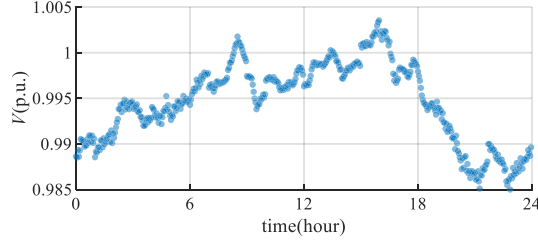


Fig. 17. Node voltages in case 1.

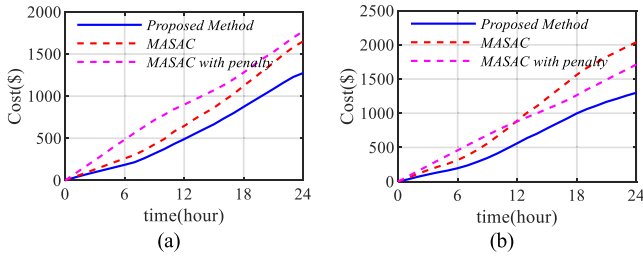


Fig. 18. Cumulative cost for microgrids: (a) microgrid 1 and (b) microgrid 2.

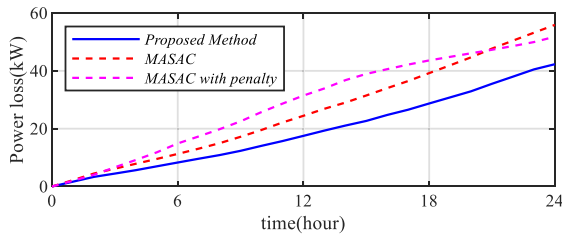


Fig. 19. Cumulative power loss for the NMG system.

since SoC of ESS in microgrid 1 is relatively low, the actual power output is negative corresponding to droop-based ESS control, which prefers to restore SoC firstly. Node voltages of the NMG system are also shown in Fig. 17, satisfying operational constraints. Further, cumulative local cost for each microgrid is presented in Fig. 18. For comparison, the results conducted by MASAC and penalty-based MASAC are also presented in Fig. 18 and the proposed method shows a better P -management performance with less generation costs. Results of cumulative power loss are plotted in Fig. 19, where the proposed method also performs better with less network loss.

To further verify the performance of droop-based PV inverter control, another case is tested. The input profiles of PV, load data

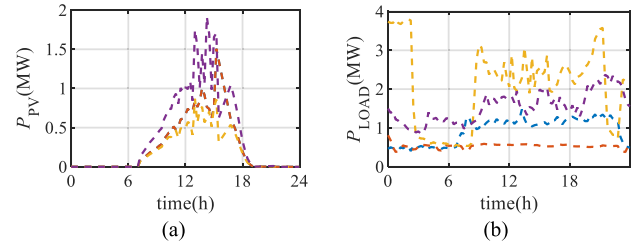


Fig. 20. Input profiles of PV, load historic data (case 2).

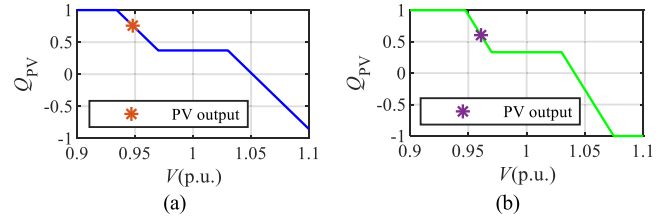


Fig. 21. Droop-based control and corresponding PV output: (a) PV inverter 1 and (b) PV inverter 4.

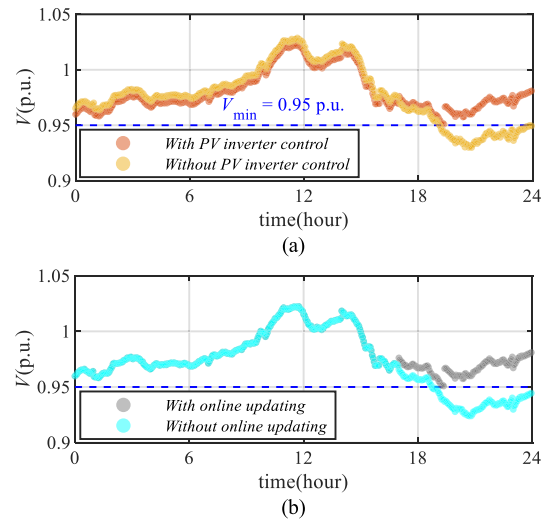


Fig. 22. Node voltages: (a) with/without PV inverter control and (b) with/without the safety model.

are illustrated in Fig. 20. PV control curves of microgrid 1, 4 and corresponding operation points are also presented in Fig. 21. If the system operates without the proposed PV control, voltage will violate the constraint. When the droop-based PV inverter control is adopted in the system, reactive power outputs will be adjusted to realize voltage regulation as shown in Fig. 22(a). Besides, the online performance of the safety model is also validated. If unlearned scenarios occur at online stage, which may lead to unsafe cases, the safety model can update the agent to ensure the safety, as presented in Fig. 22(b).

V. CONCLUSION AND FUTURE WORKS

This article developed a hierarchical and decentralized coordination in the NMG system. By utilizing historic data as

inputs, the proposed method does not require an external module for PV/load forecasting and is capable to achieve an effective operation under variable PV/load conditions. At the slow time-scale stage (1-hour), P -management agents generate the optimal setpoints for MTs and ESSs. At a faster time-scale (3-minute), Q -management agents control the outputs of PV inverters, which aims to minimize power loss and avoid voltage violations. The optimal policy is learned by a multi-agent DRL algorithm, supported by the safety model. For online application, all agents operate in a decentralized way, which only collects local information as the input. Finally, test results are given to verify the reliability and effectiveness of the proposed method.

For future works, an unbalanced power flow model can be considered to more accurately capture the network operation status. Further, battery charging/discharging degradation cost may also be included. Besides, the threshold of safety models is defined empirically in this paper. How to find the optimal threshold will also be worthy of research in the future.

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